Bangladesh University of Business & Technology (BUBT)

<u>LAB Report</u>

Course Code	: CSE 210
Course Title	: Operating Systems Lab
Experiment No.	: 04

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Intake: 52

Section: 01

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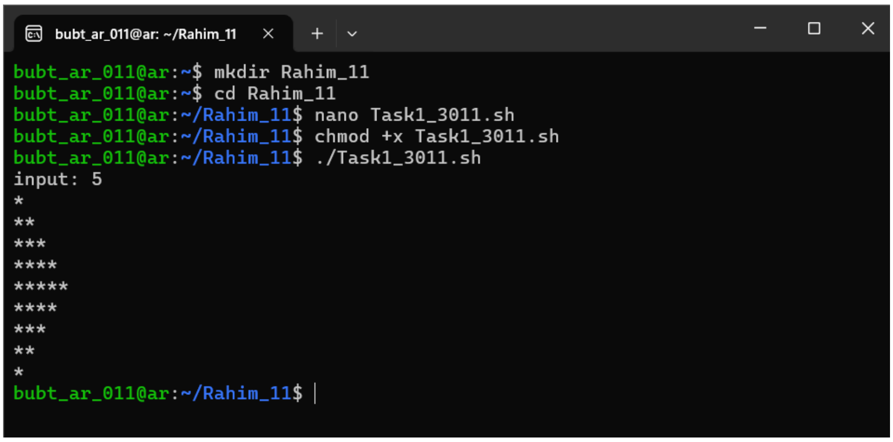
Date of Submission: 24/08/2025

Signature of Teacher

1. Print a bash script to generate the following pattern.

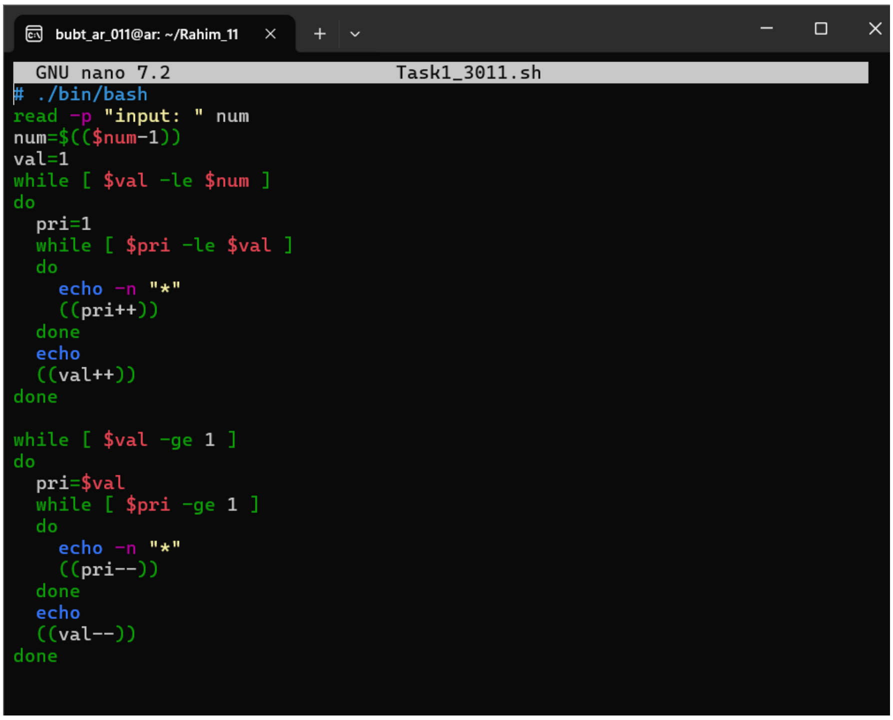
nano Task1_3011.sh
chmod +x Task1_3011.sh
./Task1_3011.sh



Inside 20234103011_Info.sh:

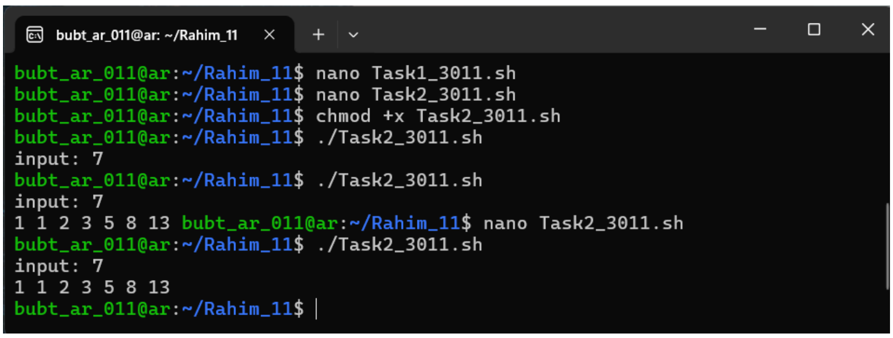
```
# ./bin/bash
read -p "input: " num
num=$(($num-1))
val=1
while [ $val -le $num ]
do
  pri=1
  while [ $pri -le $val ]
  do
    echo -n "*"
    ((pri++))
  done
  echo
  ((val++))
done

while [ $val -ge 1 ]
do
  pri=$val
  while [ $pri -ge 1 ]
  do
```

    echo -n "*"
    ((pri--))
  done
  echo
  ((val--))
done



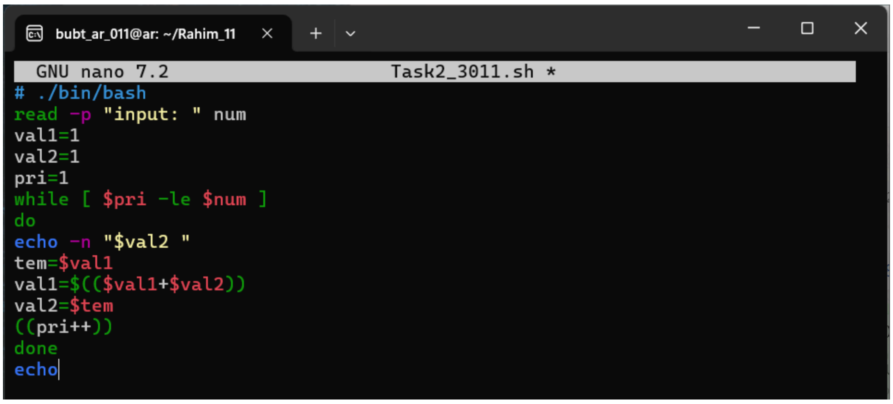
2. Print the Fibonacci series in terminal with a bash script.

nano Task2_3011.sh
chmod +x Task2_3011.sh
./Task2_3011.sh



Inside 20234103011_Operations.sh:
./bin/bash

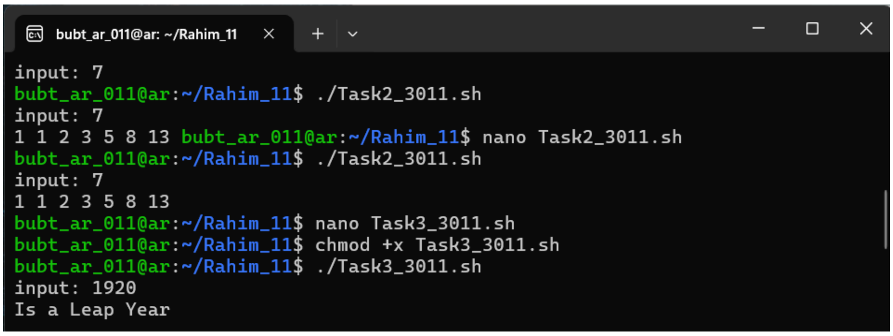
```
read -p "input: " num
val1=1
val2=1
pri=1
while [ $pri -le $num ]
do
  echo -n "$val2 "
  tem=$val1
  val1=$(($val1+$val2))
  val2=$tem
  ((pri++))
done
echo
```



```
bubt_ar_011@ar: ~/Rahim_11
GNU nano 7.2                          Task2_3011.sh *
# ./bin/bash
read -p "input: " num
val1=1
val2=1
pri=1
while [ $pri -le $num ]
do
echo -n "$val2 "
tem=$val1
val1=$(($val1+$val2))
val2=$tem
((pri++))
done
echo
```

3. Write a bash script to find if a year is a leap year or not.

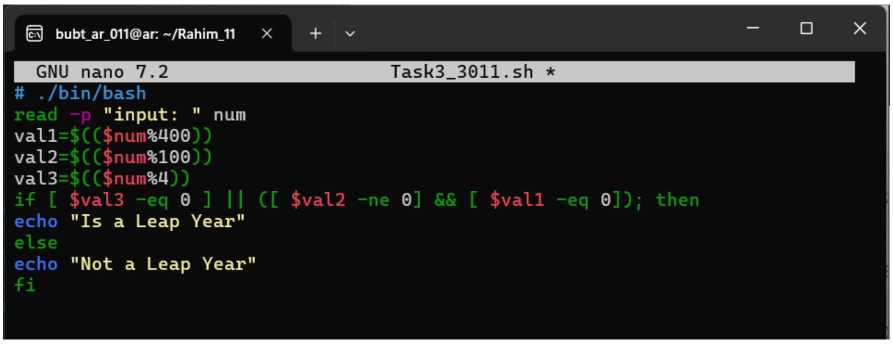
```
nano Task3_3011.sh
chmod +x Task3_3011.sh
./Task3_3011.sh
```



```
bubt_ar_011@ar: ~/Rahim_11
input: 7
bubt_ar_011@ar:~/Rahim_11$ ./Task2_3011.sh
input: 7
1 1 2 3 5 8 13 bubt_ar_011@ar:~/Rahim_11$ nano Task2_3011.sh
bubt_ar_011@ar:~/Rahim_11$ ./Task2_3011.sh
input: 7
1 1 2 3 5 8 13
bubt_ar_011@ar:~/Rahim_11$ nano Task3_3011.sh
bubt_ar_011@ar:~/Rahim_11$ chmod +x Task3_3011.sh
bubt_ar_011@ar:~/Rahim_11$ ./Task3_3011.sh
input: 1920
Is a Leap Year
```

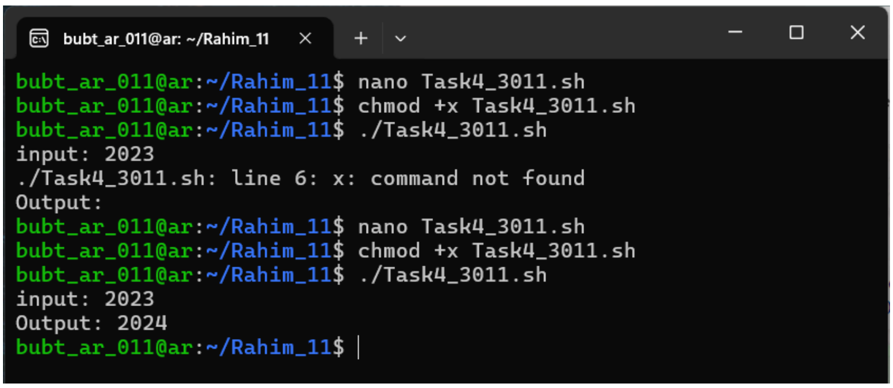
Inside 20234103011_AgeCheck.sh:

```
# ./bin/bash
read -p "input: " num
val1=$(($num%400))
val2=$(($num%100))
val3=$(($num%4))
if [ $val3 -eq 0 ] || ([ $val2 -ne 0] && [ $val1 -eq 0]); then
  echo "Is a Leap Year"
else
 echo "Not a Leap Year"
fi
```



4. Write a bash script to find the nearest leap year from the current year.

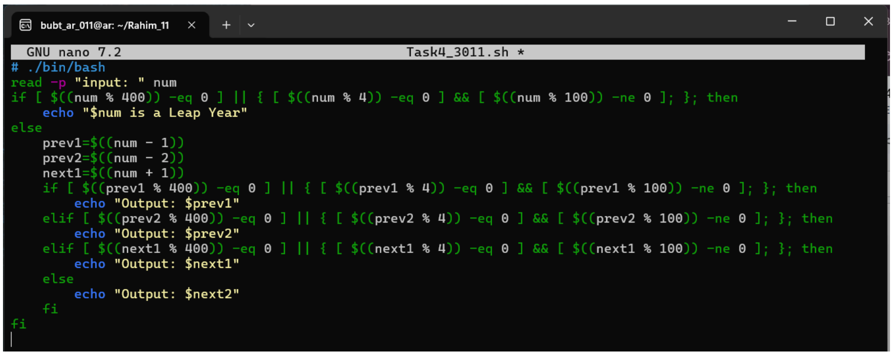
```
nano Task4_3011.sh
chmod +x Task4_3011.sh
./Task4_3011.sh
```



Inside 20234103011_ArrayAccess.sh:

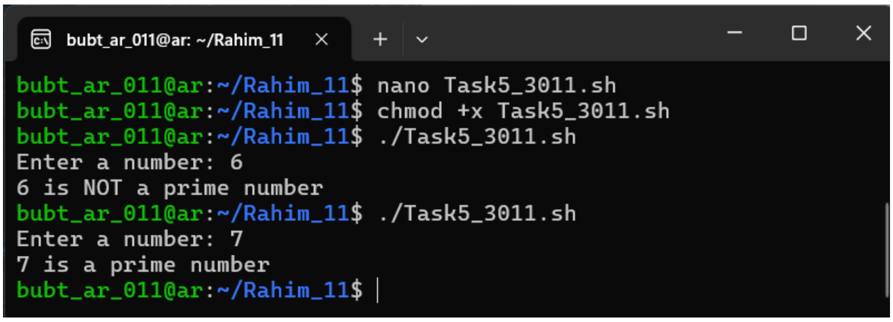
```
# ./bin/bash
read -p "input: " num
```

```
if [ $((num % 400)) -eq 0 ] || { [ $((num % 4)) -eq 0 ] && [ $((num %
            100)) -ne 0 ]; }; then
   echo "$num is a Leap Year"
else
x    prev1=$((num - 1))
   prev2=$((num - 2))
   next1=$((num + 1))
   if [ $((prev1 % 400)) -eq 0 ] || { [ $((prev1 % 4)) -eq 0 ] && [
            $((prev1 % 100)) -ne 0 ]; }; then
      echo "Output: $prev1"
   elif [ $((prev2 % 400)) -eq 0 ] || { [ $((prev2 % 4)) -eq 0 ] && [
            $((prev2 % 100)) -ne 0 ]; }; then
      echo "Output: $prev2"
   elif [ $((next1 % 400)) -eq 0 ] || { [ $((next1 % 4)) -eq 0 ] && [
            $((next1 % 100)) -ne 0 ]; }; then
      echo "Output: $next1"
   else
      echo "Output: $next2"
   fi
fi
```


```
bubt_ar_011@ar: ~/Rahim_11
GNU nano 7.2                                   Task4_3011.sh *
# ./bin/bash
read -p "input: " num
if [ $((num % 400)) -eq 0 ] || { [ $((num % 4)) -eq 0 ] && [ $((num % 100)) -ne 0 ]; }; then
    echo "$num is a Leap Year"
else
    prev1=$((num - 1))
    prev2=$((num - 2))
    next1=$((num + 1))
    if [ $((prev1 % 400)) -eq 0 ] || { [ $((prev1 % 4)) -eq 0 ] && [ $((prev1 % 100)) -ne 0 ]; }; then
        echo "Output: $prev1"
    elif [ $((prev2 % 400)) -eq 0 ] || { [ $((prev2 % 4)) -eq 0 ] && [ $((prev2 % 100)) -ne 0 ]; }; then
        echo "Output: $prev2"
    elif [ $((next1 % 400)) -eq 0 ] || { [ $((next1 % 4)) -eq 0 ] && [ $((next1 % 100)) -ne 0 ]; }; then
        echo "Output: $next1"
    else
        echo "Output: $next2"
    fi
fi
```

5. Write a bash script to check whether a given number is a prime number..

```
nano Task5_3011.sh
chmod +x Task5_3011.sh
./Task5_3011.sh
```



Inside 20234103011_ArrayAccess.sh:

```
# ./bin/bash
read -p "Enter a number: " num
if [ $num -le 1 ]; then
   echo "$num is not a prime number"
fi
is_prime=1
i=2
while [ $(($i*$i)) -le $num ]
do
   if [ $((num % i)) -eq 0 ]; then
      is_prime=0
   fi
   ((i++))
done
if [ $is_prime -eq 1 ]; then
   echo "$num is a prime number"
else
   echo "$num is NOT a prime number"
fi
```

```
bubt_ar_011@ar: ~/Rahim_11
  GNU nano 7.2                  Task5_3011.sh *
# ./bin/bash
read -p "Enter a number: " num
if [ $num -le 1 ]; then
    echo "$num is not a prime number"
fi
is_prime=1
i=2
while [ $(($i*$i)) -le $num ]
do
    if [ $((num % i)) -eq 0 ]; then
        is_prime=0
    fi
    ((i++))
done
if [ $is_prime -eq 1 ]; then
    echo "$num is a prime number"
else
    echo "$num is NOT a prime number"
fi
```

6. Write a bash script to find and print all prime numbers between two given numbers (inclusive)..

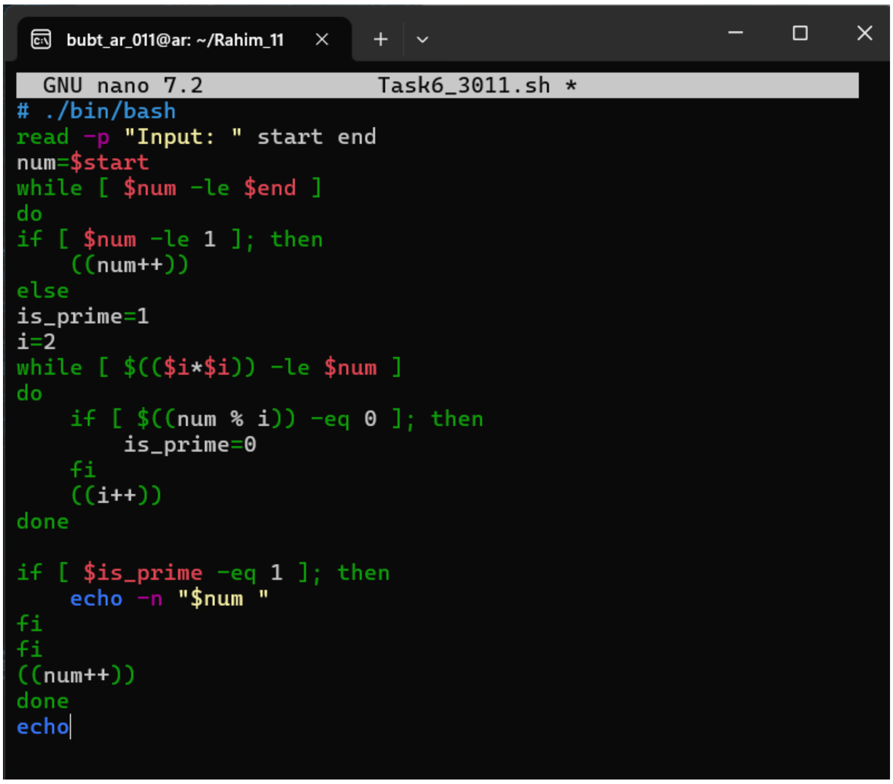
nano Task6_3011.sh
chmod +x Task6_3011.sh
./Task6_3011.sh

```
bubt_ar_011@ar: ~/Rahim_11
Enter a number: 7
7 is a prime number
bubt_ar_011@ar:~/Rahim_11$ nano Task6_3011.sh
bubt_ar_011@ar:~/Rahim_11$ chmod +x Task6_3011.sh
bubt_ar_011@ar:~/Rahim_11$ ./Task6_3011.sh
Input: 11 30
11 13 17 19 23 29
bubt_ar_011@ar:~/Rahim_11$ |
```

Inside 20234103011_ArrayAccess.sh:

```
# ./bin/bash
read -p "Input: " start end
num=$start
while [ $num -le $end ]
do
if [ $num -le 1 ]; then
   ((num++))
else
is_prime=1
```

```
i=2
while [ $(($i*$i)) -le $num ]
do
    if [ $((num % i)) -eq 0 ]; then
        is_prime=0
    fi
    ((i++))
done

if [ $is_prime -eq 1 ]; then
    echo -n "$num "
fi
fi
((num++))
done
echo
```



```
bubt_ar_011@ar: ~/Rahim_11
  GNU nano 7.2                    Task6_3011.sh *
# ./bin/bash
read -p "Input: " start end
num=$start
while [ $num -le $end ]
do
if [ $num -le 1 ]; then
    ((num++))
else
is_prime=1
i=2
while [ $(($i*$i)) -le $num ]
do
    if [ $((num % i)) -eq 0 ]; then
        is_prime=0
    fi
    ((i++))
done

if [ $is_prime -eq 1 ]; then
    echo -n "$num "
fi
fi
((num++))
done
echo
```